

Level 1: Fly in Simulation

This section documents the initial setup, system integration, and execution of Level 1 for the BeRAM AI/ML Research Assignment . The primary objective is to verify that ArduPilot SITL and Gazebo Harmonic run seamlessly together, allowing the quadcopter to arm, take off, execute flight maneuvers, and land in a 3D simulated environment .

1. Environment Setup & Dependencies

The simulation stack was deployed using WSL2 on Windows, ensuring compatibility with Linux-native ROS 2 packages and ArduPilot tools . The core components of the setup include:

- **Operating System:** Ubuntu 22.04 LTS running via WSL2 .
- **Middleware:** ROS 2 Humble to bridge communications between the simulator, autopilot, and control nodes .
- **Autopilot Software:** ArduPilot SITL (ArduCopter firmware) running as the simulated flight controller .
- **3D Simulator:** Gazebo Harmonic, rendering the vehicle physics and 3D environment .

2. Execution & Flight Verification

The system was verified by launching ArduPilot SITL alongside Gazebo and the MAVProxy console interface. As demonstrated in the simulation session, the quadcopter successfully executed the following sequence:

- **Initialization & Pre-arm Checks:** SITL, Gazebo Sim, and MAVProxy consoles established active TCP connections, loading parameter defaults and verifying EKF3 IMU alignment.
- **Mode Selection & Arming:** Switched control mode to **GUIDED**, armed the throttle/motors, and verified the **AP: Arming motors** system status message.
- **Autonomous Takeoff & Flight:** Issued the command *takeoff 5* to climb autonomously to 5 meters. The quadcopter (*iris_with_gimbal*) climbed smoothly into the 3D Gazebo world, maintaining telemetry and holding altitude accurately.

